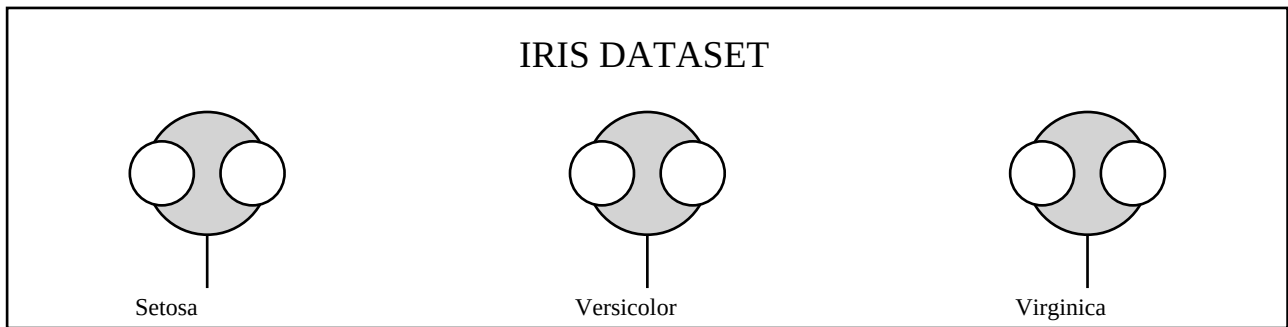


Iris Dataset & Random Forest Classifier

A beginner-friendly explanation of the code, dataset, classifier, and prediction flow.

1. What is load_iris()?

load_iris() is a built-in function from Scikit-learn (**sklearn**). It loads the famous Iris flower dataset, which is commonly used to learn and demonstrate machine learning.



The dataset contains **150 flowers** divided into 3 species: Setosa, Versicolor, and Virginica.

2. The Four Features

Feature	What it represents
Sepal length	Length of the sepal
Sepal width	Width of the sepal
Petal length	Length of the petal
Petal width	Width of the petal

Example of the Dataset

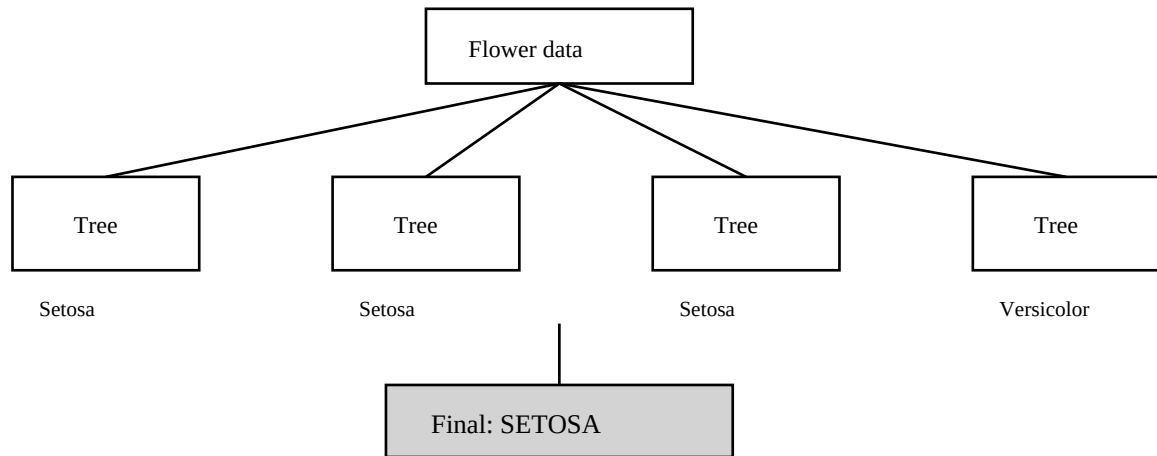
Sepal L.	Sepal W.	Petal L.	Petal W.	Species
5.1	3.5	1.4	0.2	Setosa
7.0	3.2	4.7	1.4	Versicolor
6.3	3.3	6.0	2.5	Virginica

3. What is RandomForestClassifier?

RandomForestClassifier is a supervised machine-learning algorithm used for **classification**. Classification means predicting which category or class an input belongs to.

A Random Forest combines many **Decision Trees**. Each tree makes its own prediction, and the forest uses the majority vote as the final prediction.

RANDOM FOREST



Example: if 3 trees predict **Setosa** and 1 predicts **Versicolor**, the Random Forest predicts **Setosa**.

4. Understanding Your Code

```
import streamlit as st
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.ensemble import RandomForestClassifier
```

```
def load_data():
    iris = load_iris()

    df = pd.DataFrame(iris.data,
                      columns=iris.feature_names)
    df['species'] = iris.target

    return df, iris.target_names
```

Code / Object	Purpose
<code>load_iris()</code>	Loads the Iris dataset.
<code>iris.data</code>	Contains the four numerical measurements for every flower.
<code>iris.feature_names</code>	Contains the names of the four input features.
<code>iris.target</code>	Contains the correct class labels as numbers: 0, 1, and 2.
<code>iris.target_names</code>	Contains the actual species names: Setosa, Versicolor, and Virginica.
<code>pd.DataFrame()</code>	Converts the data into a Pandas DataFrame.

5. Training the Model

```
iris = load_iris()

X = iris.data
y = iris.target

model = RandomForestClassifier(random_state=42)

model.fit(X, y)

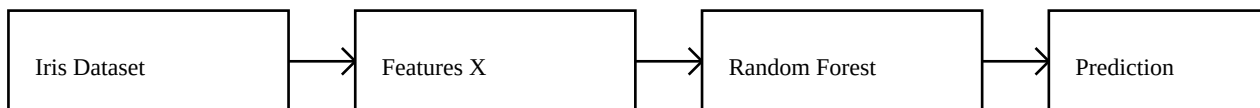
prediction = model.predict(
    [[5.1, 3.5, 1.4, 0.2]]
)

print(iris.target_names[prediction[0]])
```

What happens here?

Step	Code	Meaning
1	X = iris.data	Input features - the flower measurements.
2	y = iris.target	Correct answers - the species labels.
3	RandomForestClassifier()	Creates the machine-learning model.
4	model.fit(X, y)	Trains the model using the examples.
5	model.predict(...)	Uses the trained model to predict a new flower.

6. The Complete ML Flow



Train model with known examples

In one sentence: `load_iris()` gives you the training data, `RandomForestClassifier` learns patterns from that data, and `predict()` uses those patterns to classify a new flower.

7. The Important Mental Model

Data -> Features + Labels -> Model Training -> Prediction

Think of it like teaching a student. You show the model many flowers along with their correct species. The model learns patterns from the measurements. Later, when you give it a new flower's measurements, it predicts the species.